

Q1 TSP Algorithm Flow

Input: Sparse Distance Matrix (48 edges, 14 locations)



Step 1: Floyd-Warshall -> Complete Graph (182 paths)



Step 2: Multi-start Nearest Neighbor (14 starts)



Step 3: 2-opt Local Search (edge swap optimization)



Step 4: Simulated Annealing (T=1000, cooling=0.995)



Compare: NN+2opt vs SA -> Select Best



Output: Optimal Tour = 582km
Time = 11.64h @ 50km/h
Route: 4->6->3->2->5->...



Validation:
[OK] All 14 locations visited
[OK] Circuit closed
[OK] 582km matches official answer